

COLLEGE ADMISSION & COURSE ALLOCATION SYSTEM REPORT

1. ABSTRACT

This project focuses on developing a College Admission and Course Allocation System using Python and MySQL database. The system is designed to manage student admissions and allocate courses based on merit, preferences, and seat availability.

The main objective of this project is to automate the admission process and ensure fair allocation of courses. The system processes student data, evaluates eligibility based on marks, and assigns courses through multiple allocation rounds.

This project demonstrates how Python, database management, and automation can simplify complex admission procedures.

2. INTRODUCTION

The college admission process is often complex and time-consuming, involving multiple criteria such as marks, preferences, and seat availability. Manual processes can lead to errors and inefficiencies.

This project aims to develop a system that automates student admissions and course allocation. It ensures that students are allocated courses fairly based on their merit and preferences.

The system uses MySQL database for storing student, course, and allocation data, and provides additional features like PDF generation and CSV export.

3. OBJECTIVES

- To automate the admission process
 - To allocate courses based on merit and preferences
 - To manage student and course data
 - To generate allocation reports
 - To ensure fair and transparent allocation
 - To provide export and reporting features
-

4. SYSTEM IMPLEMENTATION

The system is implemented using Python and MySQL with the following components:

4.1 Database Connection

- Connects to MySQL database **college_db**
- Uses mysql.connector for operations

4.2 Admin Authentication

- Secure login system for admin access

4.3 Student Management

- Add student details (name, marks, category)

4.4 Course Management

- Add courses with seat capacity and minimum marks

4.5 Preference Handling

- Students can select course preferences

4.6 Allocation System

- Multi-round allocation process
- Based on:
 - Student marks (merit)
 - Course preference
 - Seat availability

4.7 Reporting Features

- Show allocation results
- Generate cutoff list
- Export data to CSV
- Generate admission PDF

5. FUNCTIONAL MODULES

5.1 Admin Login

Authenticates admin user

5.2 Add Student

Stores student data

5.3 Add Course

Stores course details

5.4 Add Preferences

Records student course choices

5.5 Allocation

Assigns courses based on merit and availability

5.6 Show Results

Displays allocation results

5.7 Cutoff Calculation

Shows minimum marks required for each course

5.8 PDF Generator

Generates admission confirmation

5.9 CSV Export

Exports allocation data

6. WORKING FLOW

1. Admin logs into the system
2. Adds students and courses
3. Students' preferences are recorded
4. Allocation process runs in rounds
5. System assigns courses based on rules
6. Results and reports are generated

7. KEY FEATURES

- Merit-based allocation system
- Multi-round admission process
- Preference-based course selection

- PDF and CSV report generation
 - Secure admin login system
-

8. ADVANTAGES

- Automates admission process
 - Reduces manual errors
 - Ensures fair allocation
 - Easy data management
 - Provides detailed reports
-

9. LIMITATIONS

- Console-based interface
 - Limited security features
 - No student login system
 - Not suitable for large-scale deployment
-

10. FUTURE ENHANCEMENTS

- Add student login portal
 - Develop web-based interface
 - Improve security features
 - Add real-time updates
 - Integrate online application system
-

11. CONCLUSION

The College Admission & Course Allocation System successfully demonstrates how Python and MySQL can be used to automate and manage the admission process efficiently.

The system ensures fair allocation based on merit and preferences while providing useful reporting features. With further enhancements, it can be developed into a full-scale online admission system.